Computational Communication Science 2
Week 2 – Lecture
Text Representation: Preprocessing, Vectorizers, and Embeddings

Anne Kroon
a.c.kroon@uva.nl

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Digital Society Minor, University of Amsterdam

Today's Agenda

Recap

Preprocessing in depth

- Tokenization
- Stemming and lemmatization

Building text representations: vectorizers

- From text to vectors
- Weighting: TF-IDF
- Pruning

Introducing embeddings

- Word embeddings

Recap

Review of Key Concepts

- **Text as data:** digital communication produces large amounts of text
- **Two approaches:**
  - *Bottom-up*: explore patterns without predefined categories
  - *Top-down*: measure predefined concepts
- **Preprocessing:** cleaning and preparing raw text
  - Tokenization, stopword removal, basic string methods

Today we go deeper into preprocessing and move on to text representation.

Preprocessing in depth

Preprocessing in depth

Tokenization

Tokenization: a reminder

Goal: split a string into a list of tokens.

```
from nltk.tokenize import TreebankWordTokenizer

docs = ["Communication in the digital society is very complex",
        "I like to study it"]

tokens = [TreebankWordTokenizer().tokenize(d) for d in docs]
print(tokens)
```

```
[['Communication', 'in', 'the', 'digital', 'society',
  'is', 'very', 'complex'],
 ['I', 'like', 'to', 'study', 'it']]
```

Result: one list of tokens per document.

What is a token?

A **token** is the basic unit of analysis after splitting text.

Tokens are usually single words, but can also be sequences of words:

Type	n	Example
Unigram	1	`"digital"`, `"society"`
Bigram	2	`"digital society"`
Trigram	3	`"the digital society"`

Why multi-word tokens? Single words sometimes lose important context.
"not good" means something very different from *"good"*.

In scikit-learn, set `ngram_range=(1,2)` in `CountVectorizer` to include both unigrams and bigrams.

Preprocessing in depth

Stemming and lemmatization

Stemming vs. lemmatization

Both reduce token forms to a common base — but differently.

Stemming

- Chops off endings using rules
- Fast, but can produce non-words
- `"society"` → `"societi"`

Lemmatization

- Maps to dictionary form
- Slower, but linguistically accurate
- `"better"` → `"good"`

Stemming in practice

Note: stemming is applied *per token* — loop over documents *and* tokens.

```
from nltk.stem import PorterStemmer

stemmer = PorterStemmer()

# tokens is a list of lists
tokens_stemmed = [[stemmer.stem(token) for token in doc]
                  for doc in tokens]
print(tokens_stemmed)
```

```
[['commun', 'in', 'the', 'digit', 'societi', 'is', 'veri', 'complex'],
 ['i', 'like', 'to', 'studi', 'it']]
```

Stemming can produce non-standard forms (`societi`, `veri`) — this is expected.

Stemming is a heuristic, not a dictionary lookup.

Lemmatization in practice: NLTK

Like stemming, lemmatization is applied *per token*.

With NLTK's `WordNetLemmatizer`, the default assumes every token is a **noun**:

```
from nltk.stem import WordNetLemmatizer

lemmatizer = WordNetLemmatizer()

words = ["studies", "geese", "running", "better"]
print([lemmatizer.lemmatize(w) for w in words])
```

```
['study', 'goose', 'running', 'better']
```

Problem: `"running"` and `"better"` are not reduced — the lemmatizer does not know they are a verb and adjective. Fixing this requires providing POS tags manually, which is impractical at scale.

Lemmatization in practice: spaCy

spaCy handles POS tagging *automatically* — no manual tags needed.

```
import spacy
nlp = spacy.load("en_core_web_sm")

doc = nlp("The studies show that running is better than doing nothing")
# show only tokens where lemma differs from the original
print([(t.text, t.lemma_) for t in doc if t.text != t.lemma_])
```

```
[('studies', 'study'), ('running', 'run'),
 ('is', 'be'), ('better', 'good'), ('doing', 'do')]
```

`"running"` → `"run"`, `"better"` → `"good"`, `"studies"` → `"study"` — all correct, automatically.

Building text representations: vectorizers

Building text representations: vectorizers

From text to vectors

A text as a bag of words

The simplest representation: count how often each token appears.

```
from collections import Counter

t = "This this is is is a test test test"
print(Counter(t.split()))
```

```
Counter({'is': 3, 'test': 3, 'This': 1, 'this': 1, 'a': 1})
```

This representation:

- preserves token frequencies
- does *not* preserve word order
- can be treated as a **vector** we can calculate with

Note: `This` and `this` are counted separately — lowercasing matters!

From vectors to a matrix

Stacking one vector per document gives a **document-term matrix (DTM)**.

```
t1 = "This this is is is a test test test"
t2 = "This is an example"
```

	a	an	example	is	This	this	test
t1	1	0	0	3	1	1	3
t2	0	1	1	1	1	0	0

Each row = document; each column = token; each cell = count.

Note: `This` and `this` are separate columns — treated as different tokens without lowercasing!

What can you do with such a matrix? Why represent texts this way?

What is a vectorizer?

- Automates building a DTM from raw text
- First **fitted** on a corpus: learns which **tokens** exist and assigns them to columns
- Can then **transform** new texts into the same feature space
- Result: a **sparse matrix** — only non-zero values stored

CountVectorizer: basic usage

```python
from sklearn.feature_extraction.text import CountVectorizer

docs = ["Communication in the digital society is very complex",
        "I like to study it"]

cv = CountVectorizer()
dtm = cv.fit_transform(docs)

print(cv.get_feature_names_out())
print(dtm.toarray())
```

```
['communication' 'complex' 'digital' 'in' 'is' 'it'
 'like' 'society' 'study' 'the' 'to' 'very']

[[1 1 1 1 1 0 0 1 0 1 0 1]
 [0 0 0 0 0 1 1 0 1 0 1 0]]
```

Note: CountVectorizer lowercases and removes punctuation by default.

Building text representations: vectorizers

Weighting: TF-IDF

Are all tokens equally informative?

Raw counts treat all tokens the same.

- `"the"` appears in almost every document — tells us very little
- `"misinformation"` in only a few documents — much more distinctive

Idea: downweight frequent tokens, upweight rare ones.

$\Rightarrow$ This is what **TF-IDF** does.

TF-IDF: the formula

$$\text{tf-idf}_{i,j} = \text{tf}_{i,j} \times \log\left(\frac{N}{\text{df}_i}\right)$$

- $\text{tf}_{i,j}$ = how often token i appears in document j
- df_i = number of documents containing token i
- N = total number of documents

Intuition:

- High TF + low DF $\rightarrow$ high score (important for this doc, rare overall)
- High DF $\rightarrow$ low score (common tokens like `"the"`)

scikit-learn applies additional normalization — see documentation.

TF-IDF vs. raw counts

Raw counts

- Interpretable
- Describes a document on its own
- Good when frequency itself matters

TF-IDF

- Adjusts for common tokens
- Better for comparing documents
- Good for search and classification

Bottom line: it is an empirical question — try both and evaluate.

Building text representations: vectorizers

Pruning

Pruning: reducing the feature space

- **Stopword removal:** fixed list, does not adapt to your corpus
- **TF-IDF:** downweights frequent tokens, but keeps them in the matrix
- **Pruning:** removes tokens based on empirical frequency in *your* corpus

Parameters in scikit-learn:

- `max_df`: drop tokens appearing in more than X% of documents
- `min_df`: drop tokens appearing in fewer than *n* documents

Putting it all together

Step 1: stopwords only

```
cv = CountVectorizer(stop_words=mystopwords)
```

Step 2: add a better tokenizer

```
cv = CountVectorizer(
    tokenizer=TreebankWordTokenizer().tokenize,
    stop_words=mystopwords)
```

Step 3: add pruning

```
cv = CountVectorizer(
    tokenizer=TreebankWordTokenizer().tokenize,
    stop_words=mystopwords,
    max_df=0.75, min_df=2)
```

Step 4: switch to TF-IDF

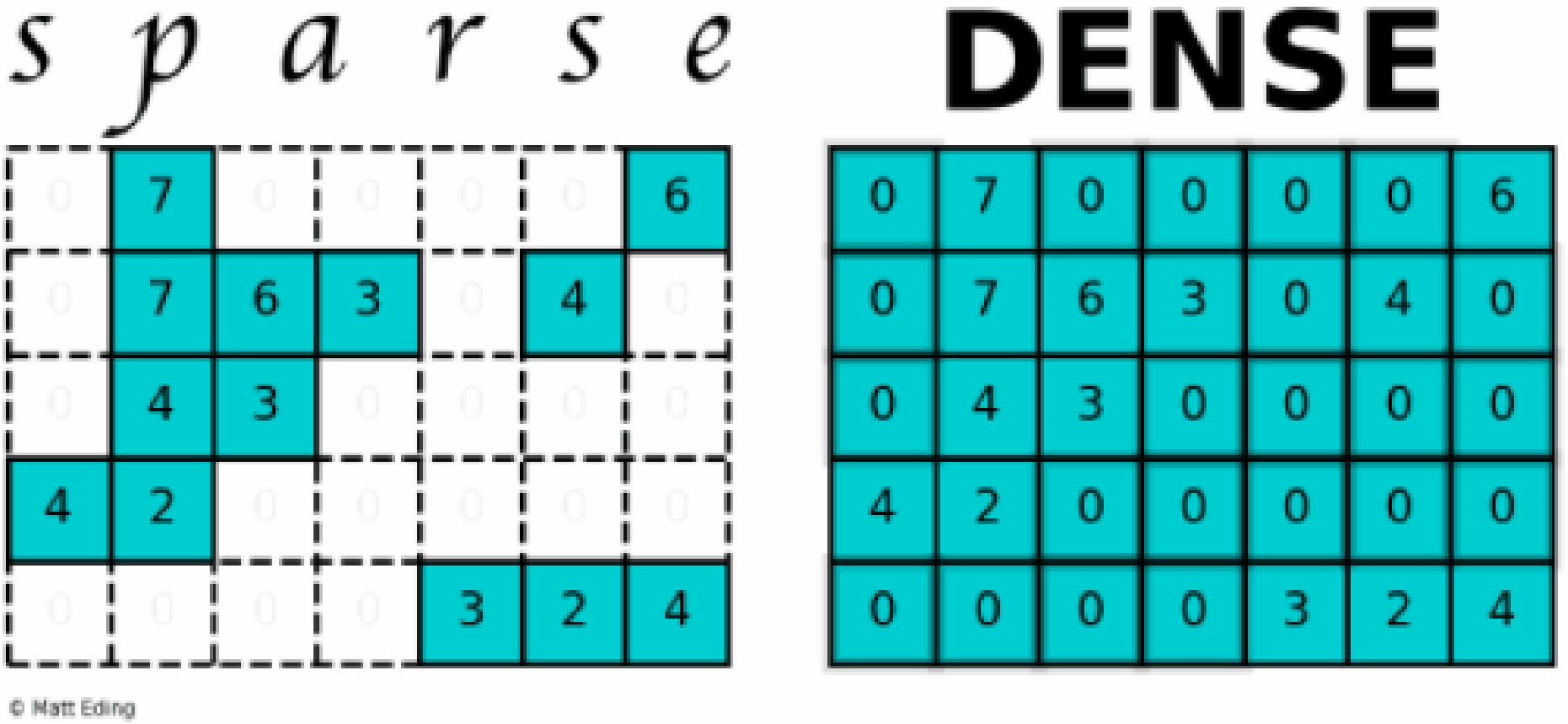
```
from sklearn.feature_extraction.text import TfidfVectorizer
tfidf = TfidfVectorizer(
    tokenizer=TreebankWordTokenizer().tokenize,
    stop_words=mystopwords,
    max_df=0.75, min_df=2)
```

Which combination is "best" — and how do you decide?

Sparse matrices

Why sparse?

- A DTM can have tens of thousands of columns (one per token)
- Most cells are zero — storing all of them wastes memory
- **Sparse matrix:** only non-zero values and positions are stored
- Never convert to dense / DataFrame on real data!



https://matteding.github.io/2019/04/25/sparse-matrices/

Explore vectorizers interactively

Best of both worlds

Use `CountVectorizer` or `TfidfVectorizer` with an NLTK-based tokenizer for full control.

Try it out yourself: DTM Explorer (Streamlit app)

Go further in code

try it out

Use `CountVectorizer` or `TfidfVectorizer` with an NLTK-based tokenizer.

Notebook: Understanding vectorizers

https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.CountVectorizer.html

Introducing embeddings

Why should communication scientists care about embeddings?

As communication researchers, we often want to know:

- Are these two news articles about the same topic?
- How similar is the language used by different politicians?
- Does the meaning of a word (e.g., *"populism"*) shift over time?

TF-IDF cannot answer these questions well — it treats every token as independent and has no notion of meaning.

Word embeddings open the door to meaning-aware text analysis.

The problem with bag-of-words

Consider these two sentences:

- *"The president gave a speech about climate change."*
- *"The leader delivered a talk on global warming."*

To a human, these sentences mean almost the same thing.
To a bag-of-words model, they share **only one token**: `"a"`.

Why? Because `"president"` and `"leader"`, `"speech"` and `"talk"`, `"climate change"` and `"global warming"` are treated as completely unrelated.

We need a representation that captures **meaning**, not just token identity.

From sparse to dense: the key idea

Bag-of-words (sparse): one dimension per token in the vocabulary.

`king = [1, 0, 0, 0, 0, 0, ..., 0]` (tens of thousands of dimensions)

Word embedding (dense): a short vector of real numbers.

`king = [0.32, −0.14, 0.87, 0.05, −0.61, ...]` (100–300 dimensions)

Each number captures some **latent aspect of meaning** — we do not define what each dimension means; the model figures it out from data.

Words used in similar contexts end up with **similar vectors**.

Introducing embeddings

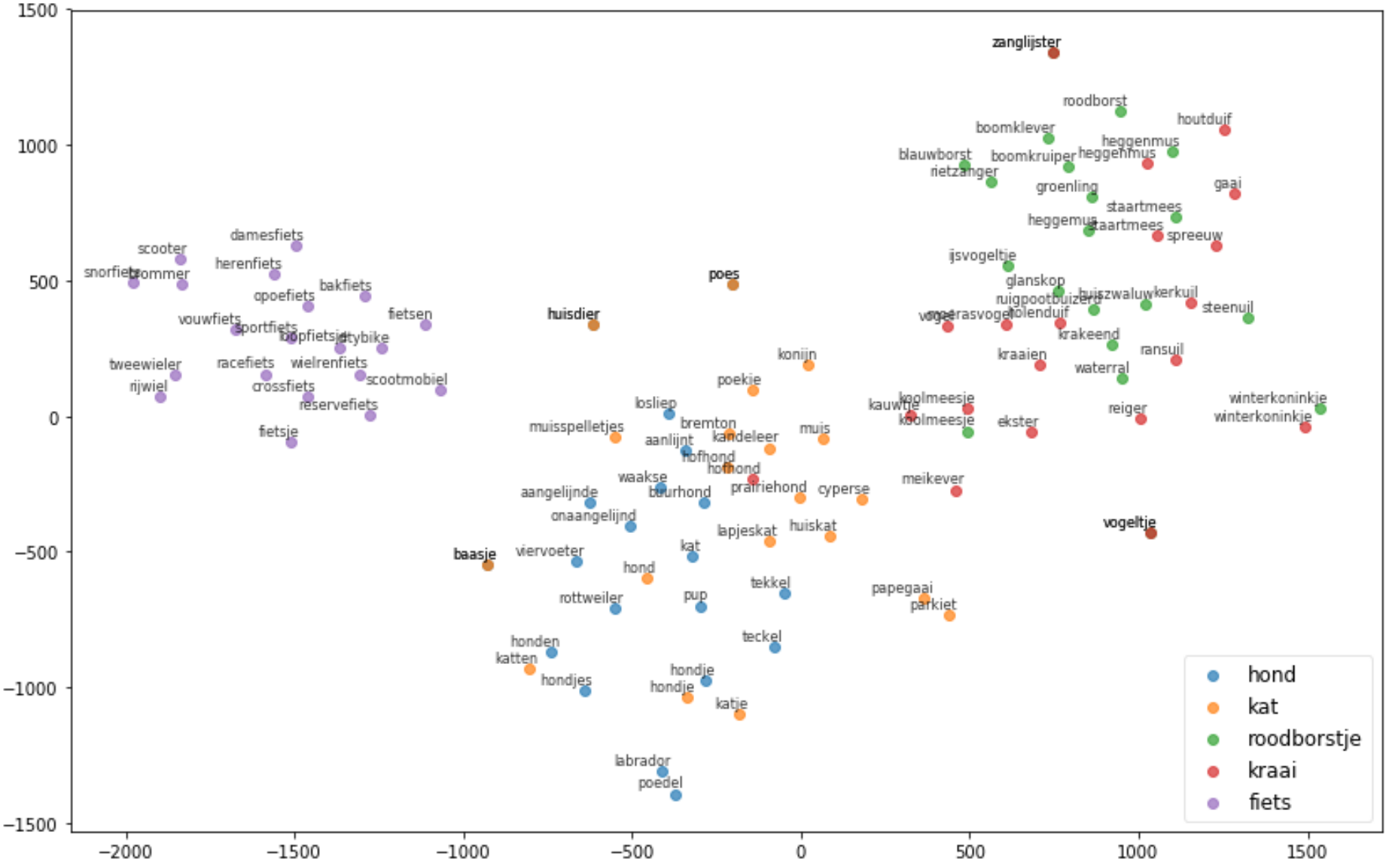
Word embeddings

How are embeddings learned?

Core idea (Firth, 1957)

"A word is characterized by the company it keeps."

- A model is trained on a large corpus (e.g., Wikipedia, news articles)
- It learns to **predict** which words appear near each other
- Words that appear in similar contexts get **similar vectors**
- No manual labelling needed — the signal comes from the text itself
- Popular models: **Word2Vec**, **GloVe**, **fastText**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opoefiets
fietsen
vouwfiets
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racefiets
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kraai
fiets

Similar words, similar vectors

Because embeddings are learned from context, words with related meanings end up **close together** in vector space.

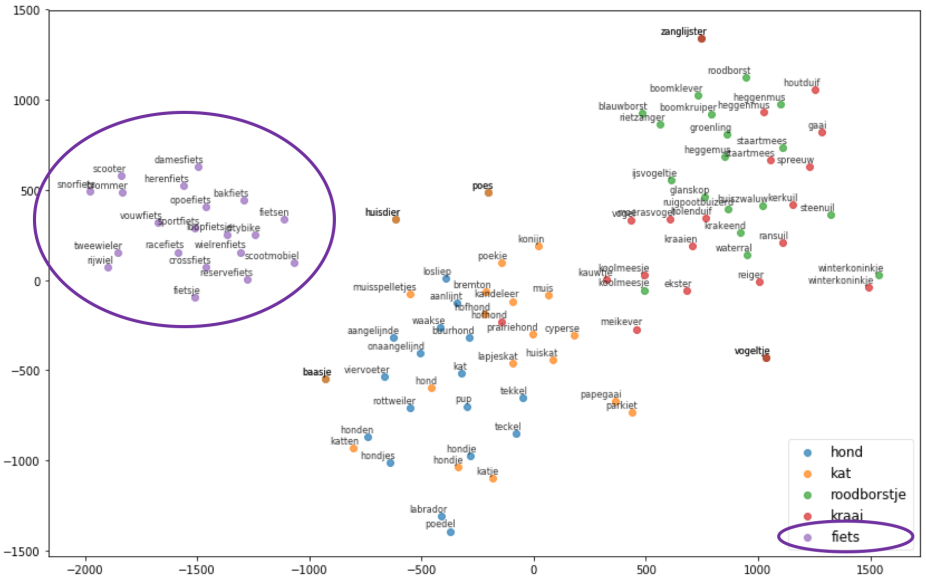
- `"dog"` and `"cat"` are close — both are pets
- `"king"` and `"queen"` are close — both are royalty
- `"happy"` and `"joyful"` are close — synonyms

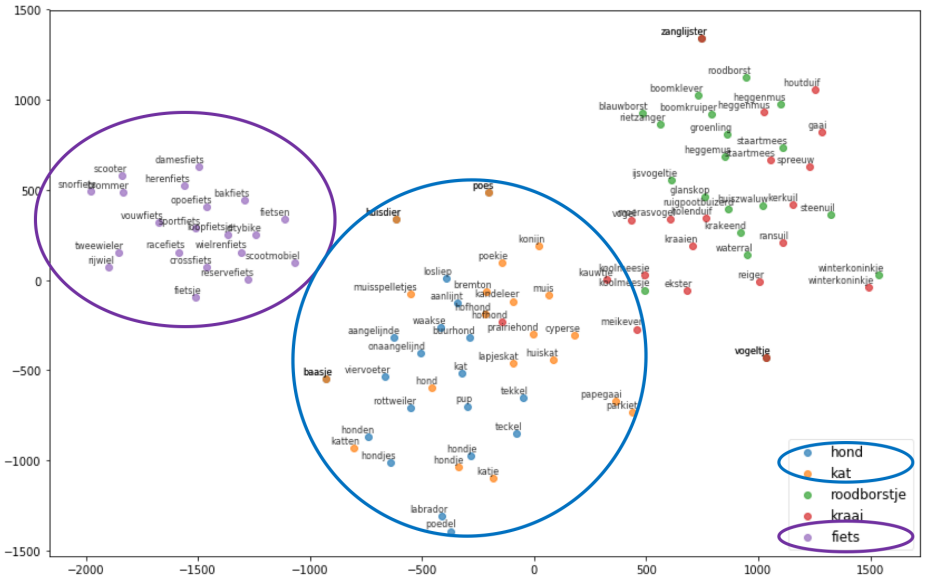
This also means you can do **arithmetic** with meaning:

$$\vec{\text{king}} - \vec{\text{man}} + \vec{\text{woman}} \approx \vec{\text{queen}}$$

The geometry of the vector space reflects real-world relationships — something TF-IDF vectors cannot do.

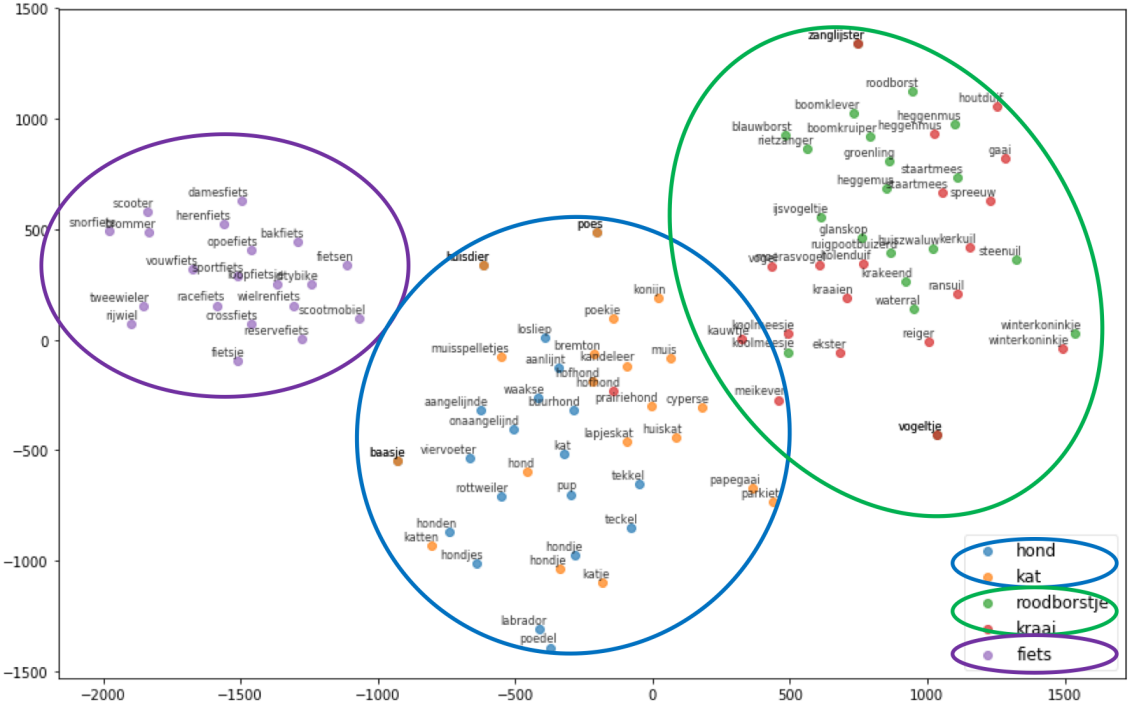
Visualizing word embeddings





Words with similar meanings cluster together in vector space.

Explore embeddings interactively



- TensorFlow Projector — explore word embeddings in 3D

Why use pretrained embeddings?

Training embeddings from scratch requires **massive amounts of text** and computing power.

Pretrained models are trained once on large corpora and shared openly:

- **Word2Vec** (Google News)
- **GloVe** (Common Crawl)
- **fastText** (subword info)
- **BERT** (contextualized)

Note: Word2Vec and GloVe give each word *one fixed vector*. BERT gives a word a *different vector depending on its context* — `"bank"` in *"river bank"* vs. *"bank account"* gets different vectors. We will return to this later.

Embeddings in communication science

What can we do with embeddings that we could not do before?

- **Sentiment analysis:** understand audience reactions beyond keyword matching
- **Recommender systems:** suggest content based on semantic similarity
- **Text similarity:** measure how alike two texts are in meaning

Note: we will work with embeddings hands-on next week.

Embeddings in practice: spaCy

Notebook: If you want to visualize your own embeddings

Wrap-up

Key takeaways

- **Preprocessing** shapes what your model sees — choices matter
- **Vectorizers** turn text into numerical matrices of token counts
- **TF-IDF** improves on raw counts by downweighting common tokens
- **Pruning** removes very rare or very frequent tokens
- **Bag-of-words ignores meaning** — synonyms and related words are unrelated features
- **Word embeddings** capture meaning and context by learning from large corpora
- There is no single best approach — it depends on your data and question

Next week

- Text similarity and distance measures
- Building a simple content-based recommender system
- Using embeddings for document comparison

Thank you!

Questions or comments?

- | a.c.kroon@uva.nl

References i